

1. Questions

In each question, three equations, I, II, and III are given. You have to solve all three equations and give the answer.

I: $7x^2 - 19x + 10 = 0$

II: $5y^2 + 17y + 14 = 0$

III: $2z^2 - 15z + 27 = 0$

a. $x > z > y$

b. $z > y > x$

c. $z > x > y$

d. $y > x > z$

e. $x > y > z$

2. Questions

I: $x^2 + 2x - 195 = 0$

II: $y^2 + 9y - 136 = 0$

III: $z^2 - 21z + 104 = 0$

Find that which of the following can't be true?

P: $x = y = z$

Q: $x > y > z$

R: $z > y > x$

a. Only P and Q

b. Only Q and R

c. Only P

d. Only R

e. None can be true

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3. Questions

I: $2x^2 - 29x + 95 = 0$

II: $2y^2 - y - 36 = 0$

III: $z^2 - 14z + 45 = 0$

Find that which of the following can't be true?

- a. $x = z$
- b. $y < z$
- c. $x > z$
- d. $x = y$
- e. $y < x$

4. Questions

A series is given below with the missing term P in it:

2, P, 154, 770, 2310, 4620

If a series is formed, whose 1st term is P, 2nd term is (1st term * 2 + 1), 3rd term is (2nd term * 2 + 2), and 4th term is (3rd term * 2 + 3), then find the 5th term of the series.

- a. 368
- b. 398
- c. 358
- d. 378
- e. 388

5. Questions

Study the following data carefully and answer the question:

Two series I and II, are given below:

I: 15, 27, 54, 99, (5P – Q), 255, 372

II: (2P – 5Q), (P – 11), 3Q, (P + 3), ?, (2P – Q)

The sum of P and Q is 45.

Find the missing term of series II.

- a. (5Q – 2)
- b. (3Q + 2)
- c. (4Q – 6)
- d. (6Q – 10)
- e. None of these

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6. Questions

A series is given below:

$(Q - 4), (P - 2Q - 2), (P - 8), (P + 2Q), (11Q + 1)$

Find the logic of the above series, if X is the n^{th} term and Y is the $(n + 1)^{\text{th}}$ term of the series.

- a. $3X - 1 = Y$
- b. $2X - 1 = Y$
- c. $2X + 1 = Y$
- d. $X + 2 = Y$
- e. $3X + 1 = Y$

7. Questions

Two series I and II are given below with the missing term P in series I and the wrong term Q in series II:

I: 15, ?, 42, 60, 81, 105

II: 138, 148, 163, 181, 202, 226

Find the ratio of P to Q.

- a. 12: 101
- b. None of these
- c. 15: 113
- d. 9: 46
- e. 3: 23

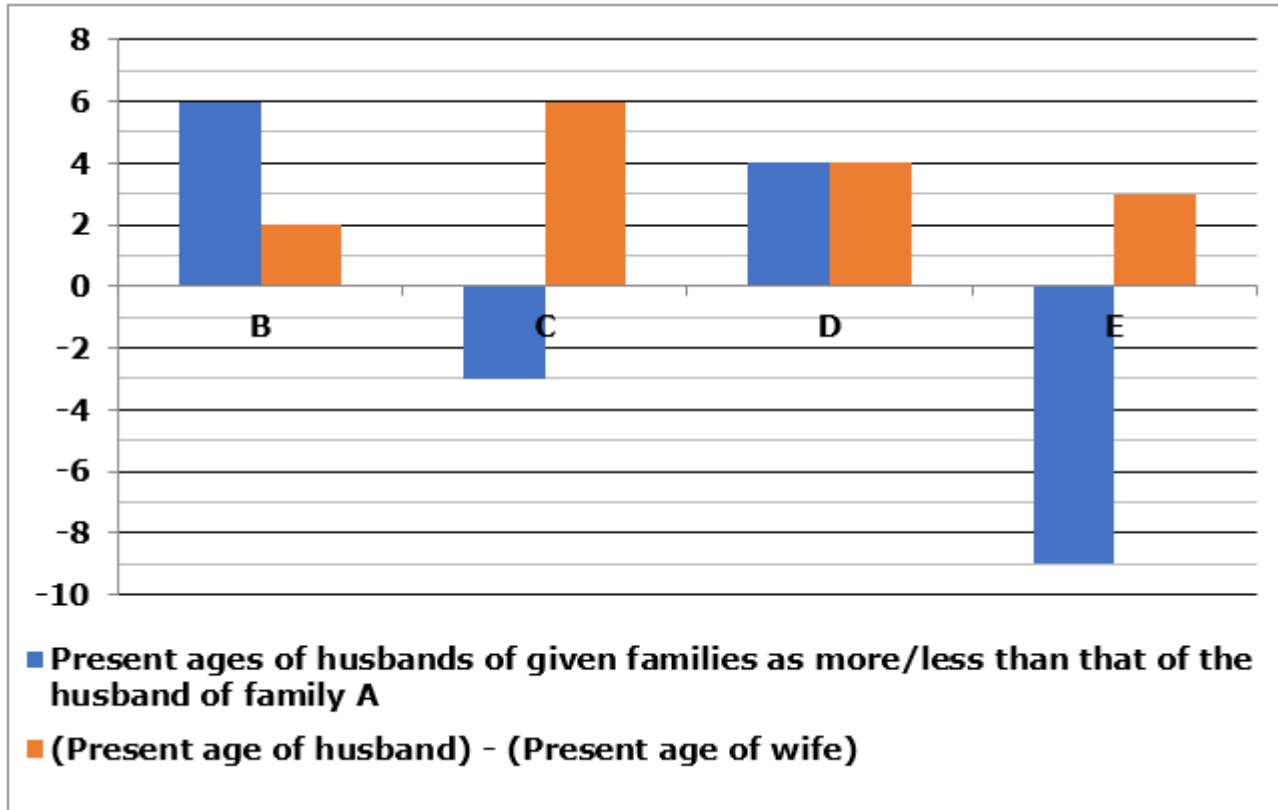
8. Questions

Study the following data carefully and answer the questions:

The data given below is related to the present ages of husbands and wives of five families: A, B, C, D, and E.

Bar graph given below shows the following data:

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Notes:

- 1:** The present ages of the husbands of families C and E are less than the present age of the husband of family A.
- 2:** The ratio of the age of wife of family C after 9 years to the age of wife of family E, 6 years ago, is 2: 1.
- There is also a child in each of families B and D. The average age of family B is 30 years, and that of family D is 29 years. Find the ratio of the age of a child of family B after 7 years to that of a child of family D after 9 years.**

- 5: 6
- 15: 19
- 3: 4
- 3: 5
- 5: 4

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9. Questions

If the ratio of husband's age in family C after 3 years to husband's age in family D after 2 years is P: Q and the difference between the present ages of wife's of families A and B is $(2P - Q)$ years, then find that the wife's present age in family A can be divisible by which of the following.

- 5
- 7
- 9

- d. Both (a) and (b)
- e. All (a), (b) and (c)

10. Questions

If a child is born in family E and after 't' years, the average age of family E becomes 23 years, then find the ratio of the husband's age of family C after 't' years to the wife's age of family C after (t – 3) years.

- a. 13: 10
- b. 12: 7
- c. 15: 11
- d. 3: 2
- e. 5: 3

11. Questions

If the present ages of husbands of families A, B, C, D, E and F are described in a circle and the degree distribution of the husband's present age of family F is 39.6° , then find the difference between the present age of husband in families B and F.

- a. 21 years
- b. 20 years
- c. 18 years
- d. 22 years
- e. 24 years

12. Questions

The age of wife in family E after 'M' years will be $58\frac{1}{3}\%$ of the age of wife in family B after 'N' years.

Find which of the following pairs of (N, M) will satisfy the condition given above?

- a. (4, 2)
- b. (6, 6)
- c. (4, 6)
- d. (8, 4)
- e. Both (b) and (d)

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13. Questions

Study the following data carefully and answer the questions:

The data given below is related to the batting average and batting strike rate of six batters A, B, C, D, E, and F in their ODI careers.

The table given below shows the following data:

Batter	Batting average	Batting strike rate
A	48	106(2/3)
B	60	90
C	40	93(1/3)
D	54	126
E	45	81
F	50	118.75

Notes:

1: Batting average of a batter = (Number of runs scored by him)/(Number of matches, in which he has been dismissed)

2: Batting strike rate of a batter = (Number of runs scored by him)/(Number of balls faced by him) * 100

If the number of matches in which C has been dismissed is 45% of his batting strike rate, then find the difference between the number of balls faced by C and the number of runs scored by C in his career?

- a. 120
- b. 105
- c. 125
- d. 112
- e. 110

14. Questions

If the number of matches, in which E has been dismissed, is equal to the difference between the batting average and batting strike rate of E and the ratio of the number of runs scored by E to F is 81: 95, then find the number of matches, in which F has been dismissed.

- a. 51
- b. 38
- c. 57
- d. 76
- e. 95

15. Questions

If the number of runs scored by F is 2850, then find which of the following statements is/are true?

- I:** The ratio of the number of runs scored by F to the number of balls faced by F is 19: 16.
- II:** The number of matches in which F has been dismissed is 21% more than his batting average.
- III:** The average of the number of runs scored by F and the number of balls faced by F is 2675.
- a. Only I
 - b. Only III
 - c. Only II
 - d. Only I and II
 - e. Only I and III

16. Questions

If the ratio of the batting average of G to the batting strike rate of G is 22: 25, then find the ratio of the number of matches in which G has been dismissed to the number of balls faced by G.

- a. Can't be determined
- b. 1: 176
- c. 1: 154
- d. 1: 88
- e. 1: 55

17. Questions

If the number of matches in which A, B, and D have been dismissed are 50, 45, and 35 respectively, then find the average of the number of balls faced by A, B, and D.

- a. 2400
- b. 2100
- c. 2250
- d. 1950
- e. 1800

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18. Questions

Study the following data carefully and answer the questions:

There are two cylindrical pits, A and B, on the ground, and both are filled with water up to some height.

The radius of pit A is 14 m, pit A is filled with the water up to a height of 20 m, and the volume of the empty part of pit A is 3080 m^3 . The difference between the depths of pits A and B is 5 m, the radius of pit B is 84% of the depth of pit A and the volume of pit B lies between 35000 m^3 and 45000 m^3 .

If the whole quantity of water of pit A is drained out and poured into pit B, then pit B will remain empty to approximately what depth?

- a. Can't be determined
- b. 16.67 m
- c. 25 m
- d. 18.89 m
- e. 21.11 m

19. Questions

What is the ratio of the depth of pit A to that of pit B?

- a. 1: 6
- b. 2: 5
- c. 7: 6
- d. 5: 6
- e. 3: 5

20. Questions

If a cuboidal pit of dimensions (25m * 24m * 20m) is dug in the ground, and all its soil is dumped into pit B, then what will be the volume of the empty part of pit B?

- a. 25850 m³
- b. 31580 m³
- c. 29580 m³
- d. 30580 m³
- e. 27850 m³

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Explanations:

1. Questions

Answer: C

From equation I:

$$\Rightarrow 7x^2 - 19x + 10 = 0$$

$$\Rightarrow 7x^2 - 14x - 5x + 10 = 0$$

$$\Rightarrow 7x(x - 2) - 5(x - 2) = 0$$

$$\Rightarrow x = 2, 5/7$$

From equation II:

$$\Rightarrow 5y^2 + 17y + 14 = 0$$

$$\Rightarrow 5y^2 + 10y + 7y + 14 = 0$$

$$\Rightarrow 5y(y + 2) + 7(y + 2) = 0$$

$$\Rightarrow y = -2, -7/5$$

From equation III:

$$\Rightarrow 2z^2 - 15z + 27 = 0$$

$$\Rightarrow 2z^2 - 6z - 9z + 27 = 0$$

$$\Rightarrow 2z(z - 3) - 9(z - 3) = 0$$

$$\Rightarrow z = 3, 9/2$$

Since both the roots of 'x' are greater than those of 'y' and both the roots of 'z' are greater than those of 'x'.

So, $z > x > y$ will be true.

2. Questions

Answer: A

From equation I:

$$\Rightarrow x^2 + 2x - 195 = 0$$

$$\Rightarrow x^2 - 13x + 15x - 195 = 0$$

$$\Rightarrow x(x - 13) + 15(x - 13) = 0$$

$$\Rightarrow x = 13, -15$$

From equation II:

$$\Rightarrow y^2 + 9y - 136 = 0$$

$$\Rightarrow y^2 + 17y - 8y - 136 = 0$$

$$\Rightarrow y(y + 17) - 8(y + 17) = 0$$

$$\Rightarrow y = 8, -17$$

From equation III:

$$\Rightarrow z^2 - 21z + 104 = 0$$

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$$\Rightarrow z^2 - 13z - 8z + 104 = 0$$

$$\Rightarrow z(z - 13) - 8(z - 13) = 0$$

$$\Rightarrow z = 13, 8$$

From P:

Since there is no common root of 'x', 'y' and 'z'.

So, $x = y = z$ can't be true.

From Q:

When $x = 13$ and $y = 8$ or -17 , then $x > y$ will be true.

Since there is no such root of 'z', which is smaller than any root of 'y'

So, $x > y > z$ can't be true.

From R:

When $z = 13$, $y = 8$, and $x = -15$, then $z > y > x$ will be true.

So, $z > y > x$ can be true.

Hence, only P and Q can't be true.

3. Questions

Answer: D

From equation I:

$$\Rightarrow 2x^2 - 29x + 95 = 0$$

$$\Rightarrow 2x^2 - 10x - 19x + 95 = 0$$

$$\Rightarrow 2x(x - 5) - 19(x - 5) = 0$$

$$\Rightarrow x = 5, 19/2$$

From equation II:

$$\Rightarrow 2y^2 - y - 36 = 0$$

$$\Rightarrow 2y^2 + 8y - 9y - 36 = 0$$

$$\Rightarrow 2y(y + 4) - 9(y + 4) = 0$$

$$\Rightarrow y = -4, 9/2$$

From equation III:

$$\Rightarrow z^2 - 14z + 45 = 0$$

$$\Rightarrow z^2 - 9z - 5z + 45 = 0$$

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$$\Rightarrow z(z - 9) - 5(z - 9) = 0$$

$$\Rightarrow z = 9, 5$$

When $x = 5$ and $z = 5$, then $x = z$ can be true.

When $y = 9/2$ or -4 and $z = 5$ or 9 , then $y < z$ can be true.

When $x = 19/2$ and $z = 5$ or 9 , then $x > z$ can be true.

Since no root of 'x' is equal to any root of 'y', so $x = y$ can't be true.

When $x = 19/2$ or 5 and $y = 9/2, -4$, then $y < x$ can be true.

Hence, option (d) is not correct.

4. Questions

Answer: D

Logic in the given series:

$$\Rightarrow 2 * 11 = 22$$

$$\Rightarrow 22 * 7 = 154$$

$$\Rightarrow 154 * 5 = 770$$

$$\Rightarrow 770 * 3 = 2310$$

$$\Rightarrow 2310 * 2 = 4620$$

So, the missing term of the series = $P = 22$

Now, the 1st term of the new series = 22

The 2nd term of the new series = $22 * 2 + 1 = 45$

The 3rd term of the new series = $45 * 2 + 2 = 92$

The 4th term of the new series = $92 * 2 + 3 = 187$

So, the 5th term of the new series = $187 * 2 + 4 = 378$

5. Questions

Answer: A

Logic in series I:

15	27	54	99	165	255	372
	+12	+27	+45	+66	+90	+117
		+15	+18	+21	+24	+27

So,

$$\Rightarrow 5P - Q = 165 \text{ -----(1)}$$

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Since the sum of P and Q = 45

So,

$$\Rightarrow P + Q = 45$$

$$\Rightarrow Q = 45 - P \text{ -----(2)}$$

From equations (1) and (2):

$$\Rightarrow 5P - 45 + P = 165$$

$$\Rightarrow P = 35$$

From equation (2):

$$\Rightarrow Q = 45 - 35$$

$$\Rightarrow Q = 10$$

Now, series II:

$$\Rightarrow (2P - 5Q), (P - 11), 3Q, (P + 3), ?, (2P - Q)$$

$$\Rightarrow (2 * 35 - 5 * 10), (35 - 11), (3 * 10), (35 + 3), ?, (2 * 35 - 10)$$

$$\Rightarrow 20, 24, 30, 38, ?, 60$$

Logic in series II:

$$\Rightarrow 20 + 4 = 24$$

$$\Rightarrow 24 + 6 = 30$$

$$\Rightarrow 30 + 8 = 38$$

$$\Rightarrow 38 + 10 = \mathbf{48}$$

$$\Rightarrow 48 + 12 = 60$$

So, the missing term of series II = 48 = (5Q - 2)

6. Questions

Answer: C

Since, P = 35 and Q = 10

So, the series:

$$\Rightarrow (Q - 4), (P - 2Q - 2), (P - 8), (P + 2Q), (11Q + 1)$$

$$\Rightarrow (10 - 4), (35 - 20 - 2), (35 - 8), (35 + 20), (110 + 1)$$

$$\Rightarrow 6, 13, 27, 55, 111$$

Logic in series:

$$\Rightarrow 6 * 2 + 1 = 13$$

$$\Rightarrow 13 * 2 + 1 = 27$$

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$$\Rightarrow 27 * 2 + 1 = 55$$

$$\Rightarrow 55 * 2 + 1 = 111$$

Since, the previous term of the series is X and the next term of the series is Y.

So, the logic in the series:

$$\Rightarrow X * 2 + 1 = Y$$

$$\Rightarrow 2X + 1 = Y$$

7. Questions

Answer: D

Logic in series I:

$$\Rightarrow 15 + 12 = \mathbf{27}$$

$$\Rightarrow 27 + 15 = 42$$

$$\Rightarrow 42 + 18 = 60$$

$$\Rightarrow 60 + 21 = 81$$

$$\Rightarrow 81 + 24 = 105$$

So, the missing term of series I = P = 27

Logic in series II:

$$\Rightarrow \mathbf{136(138)} + 12 = 148$$

$$\Rightarrow 148 + 15 = 163$$

$$\Rightarrow 163 + 18 = 181$$

$$\Rightarrow 181 + 21 = 202$$

$$\Rightarrow 202 + 24 = 226$$

So, the wrong term of series II = Q = 138

Now, the ratio of P to Q = 27: 138 = 9: 46

8. Questions

Let the present age of husband of family A = 'x' years

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Family	Present age of husband (in years)	Present age of wife (in years)
A	X	
B	$(x + 6)$	$(x + 6 - 2) = (x + 4)$
C	$(x - 3)$	$(x - 3 - 6) = (x - 9)$
D	$(x + 4)$	$(x + 4 - 4) = x$
E	$(x - 9)$	$(x - 9 - 3) = (x - 12)$

Since, the ratio of the age of wife of family C after 9 years to the age of wife of family E, 6 years ago, is 2: 1.

So,

$$\Rightarrow (x - 9 + 9)/(x - 12 - 6) = 2/1$$

$$\Rightarrow x/(x - 18) = 2/1$$

$$\Rightarrow x = 2x - 36$$

$$\Rightarrow x = 36$$

Family	Present age of husband (in years)	Present age of wife (in years)
A	36	
B	$(36 + 6) = 42$	$(36 + 4) = 40$
C	$(36 - 3) = 33$	$(36 - 9) = 27$
D	$(36 + 4) = 40$	36
E	$(36 - 9) = 27$	$(36 - 12) = 24$

Answer: C

Since, the average age of family B = 30 years

So, the present age of child in family B = $(3 * 30) - (42 + 40) = 8$ years

Since, the average age of family D = 29 years

So, the present age of child in family D = $(3 * 29) - (40 + 36) = 11$ years

Since, the child's age in family B after 7 years = $8 + 7 = 15$ years

And the child's age in family D after 9 years = $11 + 9 = 20$ years

So, the required ratio = 15: 20 = 3: 4

9. Questions

Let the present age of husband of family A = 'x' years

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Family	Present age of husband (in years)	Present age of wife (in years)
A	X	
B	$(x + 6)$	$(x + 6 - 2) = (x + 4)$
C	$(x - 3)$	$(x - 3 - 6) = (x - 9)$
D	$(x + 4)$	$(x + 4 - 4) = x$
E	$(x - 9)$	$(x - 9 - 3) = (x - 12)$

Since, the ratio of the age of wife of family C after 9 years to the age of wife of family E, 6 years ago, is 2: 1.

So,

$$\Rightarrow (x - 9 + 9)/(x - 12 - 6) = 2/1$$

$$\Rightarrow x/(x - 18) = 2/1$$

$$\Rightarrow x = 2x - 36$$

$$\Rightarrow x = 36$$

Family	Present age of husband (in years)	Present age of wife (in years)
A	36	
B	$(36 + 6) = 42$	$(36 + 4) = 40$
C	$(36 - 3) = 33$	$(36 - 9) = 27$
D	$(36 + 4) = 40$	36
E	$(36 - 9) = 27$	$(36 - 12) = 24$

Answer: E

Since, the present age of husband in family C = 33 years

And the present age of husband in family D = 40 years

$$\text{So, P: Q} = (33 + 3): (40 + 2)$$

$$\Rightarrow \text{P: Q} = 36: 42$$

$$\Rightarrow \text{P: Q} = 6: 7$$

Since, the difference between the present ages of wife of families A and B = $(2P - Q) = (2 * 6 - 7) = 5$ years

And the present age of wife in family B = 40 years

So, the present age of wife in family A will either be $40 + 5 = 45$ years and $40 - 5 = 35$ years

If the present age of wife in family A is 45 years, then it will be divisible by 5 and 9.

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If the present age of wife in family A is 35 years, then it will be divisible by 5 and 7.

So, The present age of wife in family A can be divisible by 5, 7 and 9.

10. Questions

Let the present age of husband of family A = 'x' years

Family	Present age of husband (in years)	Present age of wife (in years)
A	X	
B	(x + 6)	(x + 6 - 2) = (x + 4)
C	(x - 3)	(x - 3 - 6) = (x - 9)
D	(x + 4)	(x + 4 - 4) = x
E	(x - 9)	(x - 9 - 3) = (x - 12)

Since, the ratio of the age of wife of family C after 9 years to the age of wife of family E, 6 years ago, is 2: 1.

So,

$$\Rightarrow (x - 9 + 9)/(x - 12 - 6) = 2/1$$

$$\Rightarrow x/(x - 18) = 2/1$$

$$\Rightarrow x = 2x - 36$$

$$\Rightarrow x = 36$$

Family	Present age of husband (in years)	Present age of wife (in years)
A	36	
B	(36 + 6) = 42	(36 + 4) = 40
C	(36 - 3) = 33	(36 - 9) = 27
D	(36 + 4) = 40	36
E	(36 - 9) = 27	(36 - 12) = 24

Answer: A

The present age of husband in family E = 27 years

The present age of wife in family E = 24 years

Since a child is born in family E and after 't' years, the average age of family E becomes 23 years.

So,

$$\Rightarrow (27 + t) + (24 + t) + t = 3 * 23$$

$$\Rightarrow 3t = 18$$

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$$\Rightarrow t = 6$$

Since the present age of husband in family C = 33 years

The present age of wife in family C = 27 years

So, the required ratio = $(33 + t) : (27 + t - 3) = (33 + 6) : (27 + 6 - 3) = 39 : 30 = 13 : 10$

11. Questions

Let the present age of husband of family A = 'x' years

Family	Present age of husband (in years)	Present age of wife (in years)
A	X	
B	$(x + 6)$	$(x + 6 - 2) = (x + 4)$
C	$(x - 3)$	$(x - 3 - 6) = (x - 9)$
D	$(x + 4)$	$(x + 4 - 4) = x$
E	$(x - 9)$	$(x - 9 - 3) = (x - 12)$

Since, the ratio of the age of wife of family C after 9 years to the age of wife of family E, 6 years ago, is 2: 1.

So,

$$\Rightarrow (x - 9 + 9) / (x - 12 - 6) = 2/1$$

$$\Rightarrow x / (x - 18) = 2/1$$

$$\Rightarrow x = 2x - 36$$

$$\Rightarrow x = 36$$

Family	Present age of husband (in years)	Present age of wife (in years)
A	36	
B	$(36 + 6) = 42$	$(36 + 4) = 40$
C	$(36 - 3) = 33$	$(36 - 9) = 27$
D	$(36 + 4) = 40$	36
E	$(36 - 9) = 27$	$(36 - 12) = 24$

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Answer: B

Let the present age of husband family F = 'y' years

So, the sum of the present ages of husbands of families A, B, C, D, E and F = $(36 + 42 + 33 + 40 + 27 + y)$
 $= (178 + y)$ years

Since the degree distribution of the husband's present age of family F = 39.6°

So,

$$\Rightarrow [360/(178 + y)] * y = 39.6$$

$$\Rightarrow 360y = 7048.8 + 39.6y$$

$$\Rightarrow 320.4y = 7048.8$$

$$\Rightarrow y = 22$$

Since present age of husband's in family B = 42 years

The present age of husband's in family F = 22 years

So, the required difference = $42 - 22 = 20$ years

12. Questions

Let the present age of husband of family A = 'x' years

Family	Present age of husband (in years)	Present age of wife (in years)
A	X	
B	(x + 6)	(x + 6 - 2) = (x + 4)
C	(x - 3)	(x - 3 - 6) = (x - 9)
D	(x + 4)	(x + 4 - 4) = x
E	(x - 9)	(x - 9 - 3) = (x - 12)

Since, the ratio of the age of wife of family C after 9 years to the age of wife of family E, 6 years ago, is 2: 1.

So,

$$\Rightarrow (x - 9 + 9)/(x - 12 - 6) = 2/1$$

$$\Rightarrow x/(x - 18) = 2/1$$

$$\Rightarrow x = 2x - 36$$

$$\Rightarrow x = 36$$

Family	Present age of husband (in years)	Present age of wife (in years)
A	36	
B	(36 + 6) = 42	(36 + 4) = 40
C	(36 - 3) = 33	(36 - 9) = 27
D	(36 + 4) = 40	36
E	(36 - 9) = 27	(36 - 12) = 24

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Answer: D

The present age of wife in family B = 40 years

The present age of wife in family E = 24 years

From option (a):

The age of wife in family B after 4 years = $40 + 4 = 44$ years

The age of wife in family E after 2 years = $24 + 2 = 26$ years

Since, the required percentage = $(26/44) * 100 = 59(1/11)\%$

So, option (a) is not correct.

From option (b):

The age of wife in family B after 6 years = $40 + 6 = 46$ years

The age of wife in family E after 6 years = $24 + 6 = 30$ years

Since, the required percentage = $(30/46) * 100 = 65(5/23)\%$

So, option (b) is not correct.

From option (c):

The age of wife in family B after 4 years = $40 + 4 = 44$ years

The age of wife in family E after 6 years = $24 + 6 = 30$ years

Since, the required percentage = $(30/44) * 100 = 68(2/11)\%$

So, option (c) is not correct.

From option (d):

The age of wife in family B after 8 years = $40 + 8 = 48$ years

The age of wife in family E after 4 years = $24 + 4 = 28$ years

Since, the required percentage = $(28/48) * 100 = 58(1/3)\%$

So, option (d) is correct.

13. Questions

Answer: A

Since the batting strike rate of C = $93(1/3)$

So, the number of matches in which he has been dismissed = 45% of $93(1/3) = 42$

Since the batting average of C = (Number of runs scored by C/Number of matches, in which C has been dismissed)

So,

$\Rightarrow 40 = (\text{Number of runs scored by C}/42)$

$\Rightarrow \text{Number of runs scored by C} = 1680$

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Since the batting strike rate of C = (Number of runs scored by C/Number of balls faced by C) * 100

So,

$$\Rightarrow 93(1/3) = (1680/\text{Number of balls faced by C}) * 100$$

$$\Rightarrow 280/3 = (1680/\text{Number of balls faced by C}) * 100$$

$$\Rightarrow \text{Number of balls faced by C} = (1680 * 100) * (3/280)$$

$$\Rightarrow \text{Number of balls faced by C} = 1800$$

$$\text{Required difference} = 1800 - 1680 = 120$$

14. Questions

Answer: B

Since the batting average of E = 45

And the batting strike rate of E = 81

$$\text{So, the number of matches in which E has been dismissed} = 81 - 45 = 36$$

Since the batting average of E = (Number of runs scored by E/Number of matches, in which E has been dismissed)

So,

$$\Rightarrow 45 = (\text{No. of runs scored by E}/36)$$

$$\Rightarrow \text{No. of runs scored by E} = 1620$$

Since the ratio of the number of runs scored by E to F = 81: 95

$$\text{So, the number of runs scored by F} = 1620 * (95/81) = 1900$$

Since the batting average of F = (Number of runs scored by F/Number of matches, in which F has been dismissed)

So,

$$\Rightarrow 50 = (1900/\text{Number of matches, in which F has been dismissed})$$

$$\Rightarrow \text{Number of matches, in which F has been dismissed} = 38$$

15. Questions

Answer: A

The number of runs scored by F = 2850

Since the batting average of F = (Number of runs scored by F/Number of matches, in which F has been dismissed)

So,

$$\Rightarrow 50 = (2850/\text{Number of matches, in which F has been dismissed})$$

$$\Rightarrow \text{Number of matches, in which F has been dismissed} = 57$$

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Since the batting strike rate of F = (Number of runs scored by F/No. of balls faced by F) * 100

So,

$$\Rightarrow 118.75 = (2850/\text{Number of balls faced by F}) * 100$$

$$\Rightarrow \text{Number of balls faced by F} = (2850 * 100)/118.75$$

$$\Rightarrow \text{Number of balls faced by F} = 2400$$

From I:

Since the number of runs scored by F = 2850

And the number of balls faced by F = 2400

So, the required ratio = 2850: 2400 = 19: 16

So, I is true.

From II:

Since the number of matches in which F has been dismissed = 57

And the batting average of F = 50

$$\text{So, the required percentage} = [(57 - 50)/50] * 100 = 14\%$$

So, II is not true.

From III:

Since the number of runs scored by F = 2850

And the number of balls faced by F = 2400

$$\text{So, the required average} = (2850 + 2400)/2 = 2625$$

So, III is not true.

Hence, only I is true.

16. Questions

Answer: D

Let the batting average and batting strike rate of G be 22x and 25x respectively.

Let the number of runs scored by G be 'y'.

Let the number of matches in which G has been dismissed be P.

Let the number of balls faced by G be Q.

Since the batting average of G = (Number of runs scored by G/Number of matches, in which G has been dismissed)

So,

$$\Rightarrow 22x = (y/P)$$

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$$\Rightarrow P = (y/22x)$$

Since the batting strike rate of G = (Number of runs scored by G/Number of balls faced by G) * 100

So,

$$\Rightarrow 25x = (y/Q) * 100$$

$$\Rightarrow Q = (100y/25x)$$

$$\Rightarrow Q = (4y/x)$$

$$\text{Required ratio} = P: Q = (y/22x): (4y/x) = (1/22): 4 = 1: 88$$

17. Questions

Answer: C

Since the number of matches in which A has been dismissed = 50

And the batting average of A = (Number of runs scored by A/Number of matches, in which A has been dismissed)

So,

$$\Rightarrow 48 = (\text{Number of runs scored by A}/50)$$

$$\Rightarrow \text{Number of runs scored by A} = 2400$$

Since the batting strike rate of A = (Number of runs scored by A/Number of balls faced by A) * 100

So,

$$\Rightarrow 106(2/3) = (2400/\text{Number of balls faced by A}) * 100$$

$$\Rightarrow \text{Number of balls faced by A} = 2250$$

Since the number of matches in which B has been dismissed = 45

And the batting average of B = (Number of runs scored by B/Number of matches, in which B has been dismissed)

So,

$$\Rightarrow 60 = (\text{Number of runs scored by B}/45)$$

$$\Rightarrow \text{Number of runs scored by B} = 2700$$

Since the batting strike rate of B = (Number of runs scored by B/Number of balls faced by B) * 100

So,

$$\Rightarrow 90 = (2700/\text{No. of balls faced by B}) * 100$$

$$\Rightarrow \text{Number of balls faced by B} = 3000$$

Since the number of matches in which D has been dismissed = 35

And the batting average of D = (Number of runs scored by D/Number of matches, in which D has been dismissed)

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So,

$$\Rightarrow 54 = (\text{Number of runs scored by D}/35)$$

$$\Rightarrow \text{Number of runs scored by D} = 1890$$

$$\text{Since the batting strike rate of D} = (\text{Number of runs scored by D}/\text{Number of balls faced by D}) * 100$$

So,

$$\Rightarrow 126 = (1890/\text{Number of balls faced by D}) * 100$$

$$\Rightarrow \text{Number of balls faced by D} = 1500$$

$$\text{Required average} = (2250 + 3000 + 1500)/3 = 2250$$

18. Questions

Let the depth of pit A be 'h' m.

Since pit A is filled with water up to a height of 20 m.

So, the depth of the empty part of pit A = (h - 20) m

Since the volume of the empty part of pit A = 3080 m^3

The radius of pit A = 14 m

So,

$$\Rightarrow (22/7) * 14 * 14 * (h - 20) = 3080$$

$$\Rightarrow 22 * 2 * 14 * (h - 20) = 3080$$

$$\Rightarrow (h - 20) = 3080/(22 * 2 * 14)$$

$$\Rightarrow h - 20 = 5$$

$$\Rightarrow h = 25 \text{ m}$$

So, the depth of pit A = 25 m

Since the difference between the depths of pits A and B is 5 m.

So, the depth of pit B will either be $25 + 5 = 30 \text{ m}$ or $25 - 5 = 20 \text{ m}$.

Since the radius of pit B is 84% of the depth of pit A.

So, the radius of pit B = 84% of 25 = 21 m

Case I: When the depth of pit A is 30 m.

$$\text{The volume of pit B} = (22/7) * 21 * 21 * 30 = 41580 \text{ m}^3$$

Case II: When the depth of pit A is 20 m.

$$\text{The volume of pit B} = (22/7) * 21 * 21 * 20 = 27720 \text{ m}^3$$

Since the volume of pit B lies between 35000 m^3 and 45000 m^3 .

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So, case II can be neglected.

The depth of pit B = 30 m

Answer: E

The volume of water in pit A = $(22/7) * 14 * 14 * 20 = 12320 \text{ m}^3$

Let the height up to which pit B is filled with water be 'x' m.

Since the radius of pit B = 21 m

So,

$$\Rightarrow (22/7) * 21 * 21 * x = 12320$$

$$\Rightarrow 22 * 3 * 21 * x = 12320$$

$$\Rightarrow x = 8.89 \text{ m (approx.)}$$

Since the total depth of pit B = 30 m

So, the depth up to which pit B will remain empty = $30 - 8.89 = 21.11 \text{ m}$

19. Questions

Let the depth of pit A be 'h' m.

Since pit A is filled with water up to a height of 20 m.

So, the depth of the empty part of pit A = $(h - 20) \text{ m}$

Since the volume of the empty part of pit A = 3080 m^3

The radius of pit A = 14 m

So,

$$\Rightarrow (22/7) * 14 * 14 * (h - 20) = 3080$$

$$\Rightarrow 22 * 2 * 14 * (h - 20) = 3080$$

$$\Rightarrow (h - 20) = 3080 / (22 * 2 * 14)$$

$$\Rightarrow h - 20 = 5$$

$$\Rightarrow h = 25 \text{ m}$$

So, the depth of pit A = 25 m

Since the difference between the depths of pits A and B is 5 m.

So, the depth of pit B will either be $25 + 5 = 30 \text{ m}$ or $25 - 5 = 20 \text{ m}$.

Since the radius of pit B is 84% of the depth of pit A.

So, the radius of pit B = 84% of 25 = 21 m

Case I: When the depth of pit A is 30 m.

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The volume of pit B = $(22/7) * 21 * 21 * 30 = 41580 \text{ m}^3$

Case II: When the depth of pit A is 20 m.

The volume of pit B = $(22/7) * 21 * 21 * 20 = 27720 \text{ m}^3$

Since the volume of pit B lies between 35000 m^3 and 45000 m^3 .

So, case II can be neglected.

The depth of pit B = 30 m

Answer: D

Thus, ratio = 25: 30 = 5: 6

20. Questions

Let the depth of pit A be 'h' m.

Since pit A is filled with water up to a height of 20 m.

So, the depth of the empty part of pit A = $(h - 20)$ m

Since the volume of the empty part of pit A = 3080 m^3

The radius of pit A = 14 m

So,

$$\Rightarrow (22/7) * 14 * 14 * (h - 20) = 3080$$

$$\Rightarrow 22 * 2 * 14 * (h - 20) = 3080$$

$$\Rightarrow (h - 20) = 3080 / (22 * 2 * 14)$$

$$\Rightarrow h - 20 = 5$$

$$\Rightarrow h = 25 \text{ m}$$

So, the depth of pit A = 25 m

Since the difference between the depths of pits A and B is 5 m.

So, the depth of pit B will either be $25 + 5 = 30 \text{ m}$ or $25 - 5 = 20 \text{ m}$.

Since the radius of pit B is 84% of the depth of pit A.

So, the radius of pit B = 84% of 25 = 21 m

Case I: When the depth of pit A is 30 m.

The volume of pit B = $(22/7) * 21 * 21 * 30 = 41580 \text{ m}^3$

Case II: When the depth of pit A is 20 m.

The volume of pit B = $(22/7) * 21 * 21 * 20 = 27720 \text{ m}^3$

@QuizByMickey

Since the volume of pit B lies between 35000 m^3 and 45000 m^3 .

So, case II can be neglected.

The depth of pit B = 30 m

Answer: C

Since, the dimensions of the cuboidal pit = $25\text{m} * 24\text{m} * 20 \text{ m}$

So, the volume of the soil removed from the cuboidal pit = $25 * 24 * 20 = 12000 \text{ m}^3$

Since, the volume of pit B = $(22/7) * 21 * 21 * 30 = 41580 \text{ m}^3$

So, the volume of the empty part of pit B = $41580 - 12000 = 29580 \text{ m}^3$

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